

## Section A – (Start a new Booklet)

### Question 1. (18 marks)

Marks

(a) If  $u = 4 + 2i$  and  $v = 3 - 5i$  find:

( 4 )

(i)  $iu + \bar{v}$

(ii)  $\operatorname{Re}\left(\frac{1}{v}\right)$

(b) (i) Express each of the complex numbers  $z = 2i$  and  $w = 1 + \sqrt{3}i$  in modulus – argument form.

( 2 )

(ii) Indicate the points representing the complex numbers  $z = 2i$  and  $w = 1 + \sqrt{3}i$  on an Argand diagram.

( 2 )

(iii) Find the exact values of  $\arg\left(\frac{z}{w}\right)$  and  $\arg(z + w)$

( 4 )

(c) The polynomial equation  $x^3 - 3x^2 + x - 5 = 0$  has roots  $\alpha, \beta$  and  $\gamma$ . Find the value of  $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$

( 3 )

(d) Find  $\int x \tan^{-1} x \, dx$

( 3 )

**End of Question 1.**

**Question 2. (16 marks)**

Marks

- (a) If  $\alpha, \beta$  and  $\gamma$  are roots of  $x^3 - 4x - 1 = 0$  find the polynomial equation with roots  $2\alpha, 2\beta$  and  $2\gamma$  (answer in expanded form). ( 3 )

- (b) (i) Express  $\sqrt{8-6i}$  in the form  $a + ib$  where  $a$  and  $b$  are real and  $a > 0$ . ( 3 )

- (ii) Hence solve the equation ( 3 )

$$2z^2 + (1-3i)z - 2 = 0$$

expressing the answers in the form  $X + iY$  where  $X$  and  $Y$  are real.

- (c) Let  $A = 1 + 2i$  and  $B = 2 - i$ .

On an Argand diagram sketch the locus specified by:

- (i)  $\arg(z + B) = \frac{3\pi}{4}$  ( 2 )

- (ii)  $|z - A| = |z - B|$  ( 2 )

- (d) Find  $\int \frac{x+5}{\sqrt{x+6}} dx$  ( 3 )

**End of Question 2.**

**END OF SECTION A.**

### Section B – (Start a new Booklet)

#### Question 3. (14 marks)

Marks

(a) Find:

(i)  $\int \frac{dx}{\sqrt{3+2x-x^2}}$  ( 2 )

(ii)  $\int \frac{dx}{(x+1)(x^2+4)}$  ( 3 )

(iii)  $\int \frac{e^x + e^{2x}}{1 + e^{2x}} dx$  ( 3 )

(b) (i) Given that ( 4 )

$$u_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$$

prove that  $u_n = \frac{n}{n+1} u_{n-2}$

(ii) Evaluate  $u_5$  ( 2 )

End of Question 3.

**Question 4. (15 marks)**

Marks

- (a) If  $z = 1 + i$  is a root of the equation

$$z^3 + pz^2 + qz + 6 = 0$$

where  $p$  and  $q$  are real, find:

- (i) the values of  $p$  and  $q$  ( 3 )
- (ii) all the roots of the equation. ( 1 )
- (b) Sketch the region in the Argand diagram consisting of those points representing the complex number  $z$  which simultaneously satisfy  $|\arg z| < \frac{\pi}{3}$  and  $z + \bar{z} < 4$  and  $|z| > 2$  ( 4 )
- (c) The complex number  $z$  represents a variable point  $P$  in the Argand plane. If  $|z| = |z - \alpha|$  where  $\alpha$  is real evaluate  $\arg(z^2 - z\alpha)$  ( 3 )
- (d) Given that  $w$  is the non-real root of  $z^5 - 1 = 0$  with smallest positive argument find a real quadratic equation with roots  $w^2$  and  $w^3$ . ( 4 )

**End of Question 4.**

**END OF SECTION B.**

### Section C – (Start a new Booklet)

#### Question 5. (12 marks)

Marks

- (a) Four medical tests A, B, C and D are carried out within 14 days on a patient. A must precede B and B must precede C and D. On the days when A and B are carried out the patient must not undergo any other test, but C and D can be carried out on the same day or on different days in any order.

In how many ways can the days for the test be chosen? ( 4 )

- (b)  $K$  is a positive real number and  $u, v$  are complex numbers. Show that the points on the Argand diagram which represent respectively the numbers:

$$u, v, \frac{u - iKv}{1 - iK}$$

form the vertices of a right-angled triangle.

- (c) The quadratic equation
- $$x^2 - (2\cos\theta)x + 1 = 0$$
- has roots  $\alpha$  and  $\beta$ .

(i) Find expressions for  $\alpha$  and  $\beta$  in terms of  $\theta$  ( 2 )

(ii) Show that  $\alpha^{10} + \beta^{10} = 2\cos(10\theta)$  ( 2 )

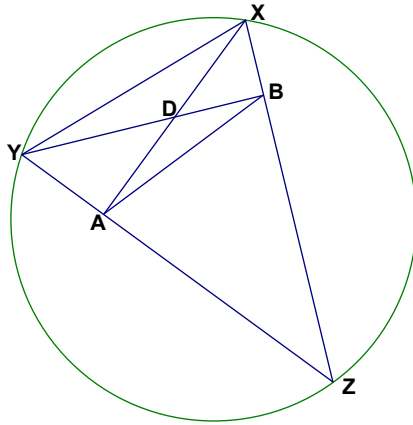
**End of Question 5.**

**Question 6. (15 marks)**

Marks

- (a) When a cubic polynomial  $A(x)$  is divided by  $x^2 + x + 1$  the remainder is  $2x + 3$ . When  $A(x)$  is divided by  $x(x + 3)$  the remainder is  $5(x + 1)$ . Find the polynomial  $A(x)$  ( 3 )

(b)



XY is a fixed chord of a circle. Z is a point on major arc XY.  
The perpendicular from X to the chord YZ meets YZ at A. The  
perpendicular from Y to the chord XZ meets XZ at B.  
Let  $\angle XZY = \theta$ .

- (i) Copy the diagram into your booklet
- (ii) Prove that ABXY is a cyclic quadrilateral. ( 1 )
- (iii) Show that  $AB = XY \cos \theta$ . ( 3 )
- (iv) Deduce that as Z moves on the major arc XY the length of AB is constant. ( 2 )
- (c) (i) Suppose that  $k$  is a double root of the polynomial equation  $P(x) = 0$ . Show that  $P'(k) = 0$  ( 2 )
- (ii) If  $x^3 + 3px^2 + 3qx + r = 0$  has a double root prove that ( 4 )  
 $(pq - r)^2 = 4(p^2 - q)(q^2 - pr)$

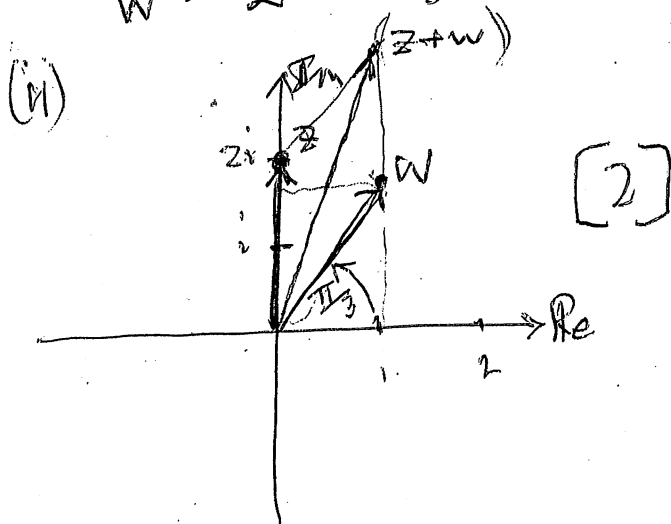
**End of Question 6.****END OF SECTION C.****END OF THE EXAMINATION.**

Question 1

$$\begin{aligned}
 (a)(i) \quad iu + \bar{v} &= i(4+2i) + (3+5i) \\
 &= -2+4i + 3+5i \\
 &= 1+9i \quad [2]
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \operatorname{Re}\left(\frac{1}{\sqrt{v}}\right) &= \operatorname{Re}\left[\frac{\bar{v}}{\sqrt{v}\bar{v}}\right] \\
 &= \operatorname{Re}\left[\frac{1}{|v|^2} \cdot \bar{v}\right] \\
 &= \frac{1}{|v|^2} \cdot \operatorname{Re}(\bar{v}) \\
 &= \frac{1}{34} \times 3 \\
 &= \frac{3}{34} \quad [2]
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad z &= 2i, \quad w = 1+\sqrt{3}i \\
 (i) \quad z &= 2\operatorname{cis}\frac{\pi}{2} \\
 w &= 2\operatorname{cis}\frac{\pi}{3} \quad [2]
 \end{aligned}$$



$$\begin{aligned}
 (iii) \quad \arg\left(\frac{z}{w}\right) &= \arg z - \arg w \\
 &= \frac{\pi}{2} - \frac{\pi}{3} \\
 &= \frac{\pi}{6} \quad [2] \\
 \arg(z+w) &= \frac{1}{2}\left(\frac{\pi}{2} + \frac{\pi}{3}\right) \\
 &= \frac{5\pi}{12} \quad [2]
 \end{aligned}$$

$$(c) \quad x^3 - 3x^2 + x - 5 = 0$$

$$\begin{aligned}
 \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\
 &= \frac{1}{5} \quad [3]
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad \int x \tan^{-1} x \, dx \\
 &= \int \frac{d}{dx}\left(\frac{x^2}{2}\right) \cdot \tan^{-1} x \, dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \cdot \frac{d}{dx} \tan^{-1} x \, dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C \\
 &= \frac{1}{2} (x^2 + 1) \tan^{-1} x - \frac{1}{2} x + C \quad [3]
 \end{aligned}$$

Question 2

$$(a) \quad x^3 - 4x - 1 = 0$$

We seek eq'n w. roots  $2x$  etc

$$\text{Let } y = 2x; \text{ So } x = \frac{y}{2}$$

Substitute:

$$\left(\frac{y}{2}\right)^3 - 4\left(\frac{y}{2}\right) - 1 = 0$$

$$\frac{y^3}{8} - 2y - 1 = 0$$

$$y^3 - 16y - 8 = 0$$

Replace  $y$  with  $x$ .

$$x^3 - 16x - 8 = 0 \quad [3]$$

Q2 (Contd)

(2)

(b)(i)  $\sqrt{8-6i} = 3-i$  (calculator)

OR

$$\sqrt{8-6i} = a+ib$$

$$8-6i = (a+ib)^2$$

$$= a^2 - b^2 + 2iab$$

$$a^2 - b^2 = 8 \quad ; \quad 2ab = -6$$

$$ab = -3$$

$$b = -\frac{3}{a}$$

$$a^2 - \left(-\frac{3}{a}\right)^2 = 8$$

$$a^4 - 9 = 8a^2$$

$$a^4 - 8a^2 - 9 = 0$$

$$(a^2 - 9)(a^2 + 1) = 0$$

$$\therefore a = \pm 3$$

$$b = \mp 1$$

[3]

$$\sqrt{8-6i} = 3-i \text{ (Principal Root)}$$

(ii)  $2z^2 + (1-3i)z - 2 = 0$

$$z = \frac{-(1-3i) \pm \sqrt{(1-3i)^2 - 4(2)(-2)}}{4}$$

$$= \frac{-1+3i \pm \sqrt{-8-6i+16}}{4}$$

$$= \frac{-1+3i \pm \sqrt{8-6i}}{4}$$

$$= \frac{-1+3i \pm (3-i)}{4}$$

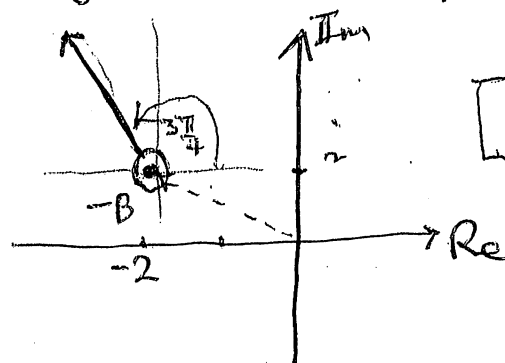
$$= \frac{1}{4}(2+2i), \frac{1}{4}(-4+4i)$$

$$= \frac{1}{2} + \frac{1}{2}i, -1+i \quad [3]$$

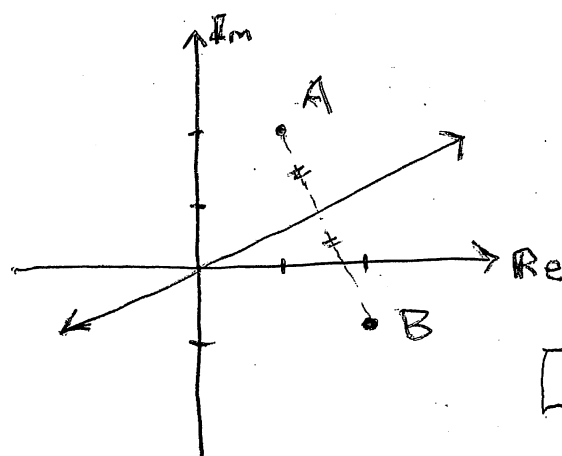
(c)  $A = 1+2i$  ;  $B = 2-i$

(i)  $\arg(z+B) = \frac{3\pi}{4}$

ie  $\arg(z - (-B)) = \frac{3\pi}{4}$



(ii)  $|z-A| = |z-B|$



(d)  $\int \frac{x+5}{\sqrt{x+6}} dx$  Let  $u = x+6$   
 $du = dx$

$$= \int \frac{u-1}{\sqrt{u}} du$$

$$= \int (u^{1/2} - u^{-1/2}) du$$

$$= \frac{u^{3/2}}{3/2} - \frac{u^{1/2}}{1/2} + C$$

$$= \frac{2}{3}(x+6)^{3/2} - 2(x+6)^{1/2} + C$$

$$= \frac{2}{3}\sqrt{x+6}\left(\frac{x}{3}+1\right) + C \quad [3]$$

# Solutions to section (B) Ex + (II) Assessment Task (1)

Question (3) (14 mks)

$$\int \frac{dx}{\sqrt{3+2x-x^2}}$$

Now,  $-(x^2-2x-3)$   
 $= -[(x-1)^2-4]$   
 by completing square

$$\int \frac{dx}{\sqrt{4-(x-1)^2}} \quad (2)$$

$$= \sin^{-1}\left(\frac{x-1}{2}\right) + c$$

(ii) let

$$\frac{1}{(x+1)(x^2+4)} = \frac{a}{x+1} + \frac{bx+c}{x^2+4}$$

$$1 = a(x^2+4) + (bx+c)(x+1)$$

put  $x = -1$   $1 = 5a$

$$\therefore a = \frac{1}{5}$$

Equate coefficients  
 of  $x^2$   $a+b=0$   
 $\Rightarrow b = -a = -\frac{1}{5}$

put  $x=0 \Rightarrow c = 1 - \frac{4}{5} = \frac{1}{5}$   
 $1 = 4a+c$

$$\frac{1}{(x+1)(x^2+4)} = \frac{1}{5(x+1)} - \frac{x}{5(x^2+4)} + \frac{1}{5(x^2+4)}$$

$$\int \frac{dx}{(x+1)(x^2+4)} = \frac{1}{5} \int \left( \frac{1}{x+1} - \frac{x}{x^2+4} + \frac{1}{x^2+4} \right) dx$$

$$= \frac{1}{5} \ln|x+1| - \frac{1}{10} \ln(x^2+4) + \frac{1}{10} \tan^{-1}\left(\frac{x}{2}\right) + c \quad (3)$$

(iii)  $\int \frac{e^x + e^{2x}}{1+e^{2x}} dx$   
 "splitting"

$$= \int \frac{e^x dx}{1+(e^x)^2} + \int \frac{e^{2x} dx}{1+e^{2x}}$$

Let  $u = e^x$   $\frac{du}{dx} = e^x$   
 $\int \frac{du}{1+u^2} = \tan^{-1}(e^x)$

$$\int \frac{e^{2x} dx}{1+e^{2x}}$$

Now, let  $u = 1+e^{2x}$   
 $du = 2e^{2x} dx$   
 $= \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln u$   
 $= \frac{1}{2} \ln(1+e^{2x})$

$$\therefore \int \frac{e^x + e^{2x}}{1+e^{2x}} dx = \tan^{-1}(e^x) + \frac{1}{2} \ln(1+e^{2x}) + c \quad (3)$$

(b) (i)  $M_n = \int_0^1 (1-x^2)^{n/2} dx$   
 From integration by parts:

$$M_n = \int_0^1 (1-x^2)^{n/2} \frac{d}{dx}(x) dx$$

$$= \left[ x(1-x^2)^{n/2} \right]_0^1 - \int_0^1 x \cdot \frac{n}{2} (1-x^2)^{n/2-1} (-2x) dx$$

$$= n \int_0^1 x^2 (1-x^2)^{n/2-1} dx$$

$$= -n \int_0^1 (1-x^2)^{n/2-1} dx$$

3(b) (i) (cont.)

$$= -n \int_0^1 (1-x^2)(1-x^2)^{n/2-1} dx$$

$$+ n \int_0^1 (1-x^2)^{n/2-1} dx$$

$$M_n = -n \int_0^1 (1-x^2)^{n/2} dx + n \int_0^1 (1-x^2)^{n/2-1} dx$$

$$\therefore M_n = -n M_n + n M_{n-2}$$

$$\therefore (n+1) M_n = n M_{n-2}$$

$$\therefore M_n = \left( \frac{n}{n+1} \right) M_{n-2}$$

(b) (ii) Now,  $M_5 = \frac{5}{6} M_3$

$$M_3 = \frac{3}{4} M_1$$

$$M_1 = \int_0^1 \sqrt{1-x^2} dx$$

$$= \frac{\pi}{4} \text{ (from diagram)}$$

$$\therefore M_3 = \frac{3\pi}{16} \Rightarrow M_5 = \frac{5 \times 3\pi}{6 \times 16}$$

$$\therefore M_5 = \frac{15\pi}{128} = \frac{5\pi}{256}$$

Question (4) (15 mks)

(a) If  $1+i$  is a root, then

$$(1+i)^3 + p(1+i)^2 + q(1+i) + 6 = 0$$

$$(1+i)^2 = 2i, (1+i)^3 = -2+2i$$

$$(q+4) + (2+2p+q)i = 0$$

$$\therefore q = -4, q+4 = 0$$

$$\text{or } 2+2p+q = 0 \quad (3)$$

$$\Rightarrow 2p = -2-q$$

$$= -2+4$$

$$= 2$$

$$\therefore p = 1, q = -4$$

$$\therefore z^3 + z^2 - 4z + 6 = 0$$

is the equation. (1)

$$\sum \alpha_i = -1$$

$$(1+i) + (1-i) + \beta = -1$$

$$\therefore 2 + \beta = -1$$

$$\Rightarrow \beta = -3$$

$$\therefore \text{roots are } 1 \pm i, -3$$

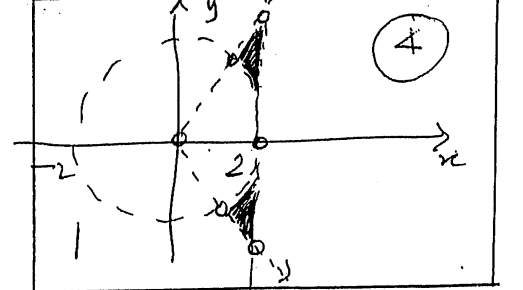
(b)  $|\arg z| < \frac{\pi}{3}$

$$-\frac{\pi}{3} < \arg z < \frac{\pi}{3}$$

$$z + \bar{z} < 4 \Rightarrow 2x < 4$$

$$x < 2$$

$$|z| > \frac{2}{\sqrt{3}}$$



$$x=2 \quad (3)$$

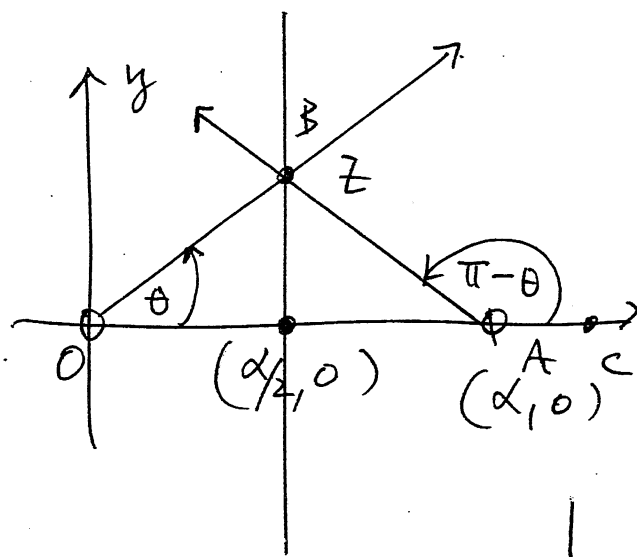
(c)  $|z| = |z-\alpha|$

$\therefore$  locus is the perp. bisector joining  $(0,0)$

and  $(\alpha,0)$ .

$$\arg(z^2 - \alpha z) = \arg[z(z-\alpha)]$$

# Question 4(c) (cont.)



$$\angle + \arg Z = \theta$$

$\therefore \triangle OAB$  is  
Isosceles.

$$\therefore \angle BAO = \theta$$

$$\therefore \angle CAB = \pi - \theta$$

$$\begin{aligned} \therefore \arg Z + \arg(Z - \alpha) \\ &= \theta + \pi - \theta \\ &= \pi. \end{aligned}$$

$$\therefore \arg(Z^2 - \alpha Z) = \pi$$

$$(d) \quad z^5 - 1 = 0$$

4

$$\therefore z^5 = 1 = \cos(2k\pi) + i \sin(2k\pi)$$

$$\therefore z = \cos\left(\frac{2k\pi}{5}\right) + i \sin\left(\frac{2k\pi}{5}\right)$$

$$k = 0, 1, 2, 3, 4.$$

$$z_1 = \cos(0) = 1$$

$$w = z_2 = \cos\frac{2\pi}{5} + i \sin\frac{2\pi}{5}$$

$$w^2 = z_3 = \cos\frac{4\pi}{5} + i \sin\frac{4\pi}{5}$$

$$w^3 = z_4 = \cos\frac{6\pi}{5} + i \sin\frac{6\pi}{5}$$

$$\begin{aligned} z &= \cos\left(2\pi - \frac{4\pi}{5}\right) \\ &= \cos\left(-\frac{4\pi}{5}\right) \\ &= \cos\frac{4\pi}{5} - i \sin\frac{4\pi}{5} \end{aligned}$$

$$\begin{aligned} w^4 = z_5 &= \cos\frac{8\pi}{5} + i \sin\frac{8\pi}{5} \\ &= \cos\left(2\pi - \frac{2\pi}{5}\right) \\ &= \cos\frac{2\pi}{5} - i \sin\frac{2\pi}{5} \end{aligned}$$

$$w^2 + w^3 = 2 \cos\frac{4\pi}{5}$$

$$w^2 \cdot w^3 = w^{2+3} = w^5 = 1$$

$$z^2 - (w^2 + w^3)z + (w^2 \cdot w^3) = 0$$

$\therefore$  real quad. equation  
whose roots are  $w^2, w^3$  is

$$z^2 - \left(2 \cos\frac{4\pi}{5}\right)z + 1 = 0$$

2009 Mathematics Extension 2 Assessment 1: **Section C** solutions

5. (a) Four medical tests  $A$ ,  $B$ ,  $C$ , and  $D$  are carried out within fourteen days on a patient.  $A$  must precede  $B$  and  $B$  must precede  $C$  and  $D$ . On the days when  $A$  and  $B$  are carried out the patient must not undergo any other tests, but  $C$  and  $D$  can be carried out on the same day or on different days in any order.

4

In how many ways can the days for the tests be chosen?

**Solution: Method 1:**

Case 1— consider when  $C$  and  $D$  are on different days. Ways of choosing the days is  $\binom{14}{4}$ . Then we know that the first day is  $A$  and the second is  $B$  so we have to order  $C$  and  $D$ , i.e.  $2!$  ways. So the number of ways if 4 different days is  $\binom{14}{4} \times 2!$ .

Case 2— consider when  $C$  and  $D$  are on the same day. Ways of choosing the days is  $\binom{14}{3}$ .

$$\therefore \text{Total ways} = \binom{14}{4} \times 2! + \binom{14}{3},$$

$$= 2366.$$

**Method 2:**

|   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| A | B |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|

If  $A$  and  $B$  are on the first 2 days,  $C$  can be in any of 12 places AND  $D$  can be in any of 12 places  $\implies$  Ways =  $12^2$ .

|  |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|
|  | A | B |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|

If  $A$  and  $B$  are on the second 2 days,  $C$  can be in any of 11 places AND  $D$  can be in any of 11 places  $\implies$  Ways =  $11^2$  ...

Hence the total number of ways with  $A$  and  $B$  together is  $12^2 + 11^2 + 10^2 + \dots + 2^2 + 1 = 650$ .

|   |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|--|---|--|--|--|--|--|--|--|--|--|--|--|--|--|
| A |  | B |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|--|---|--|--|--|--|--|--|--|--|--|--|--|--|--|

If  $A$  and  $B$  are over the first 3 days,  $C$  can be in any of 11 places AND  $D$  can be in any of 11 places  $\implies$  Ways =  $11^2$ .

|  |   |  |   |  |  |  |  |  |  |  |  |  |  |  |  |
|--|---|--|---|--|--|--|--|--|--|--|--|--|--|--|--|
|  | A |  | B |  |  |  |  |  |  |  |  |  |  |  |  |
|--|---|--|---|--|--|--|--|--|--|--|--|--|--|--|--|

If  $A$  and  $B$  are over the second 3 days,  $C$  can be in any of 10 places AND  $D$  can be in any of 10 places  $\implies$  Ways =  $10^2$  ...

Hence the total number of ways with  $A$  and  $B$  over three days is  $11^2 + 10^2 + 9^2 + \dots + 2^2 + 1 = 506$ .

$\vdots$

Therefore, the total number of ways with  $A$  and  $B$  in all possible positions is  $650 + 506 + 385 + \dots + 5 + 1 = 2366$ .

- (b)  $K$  is a positive real number and  $u, v$  are complex numbers. Show that the points on the Argand diagram which represent respectively the numbers:

$$u, v, \frac{u - iKv}{1 - iK}$$

form the vertices of a right-angled triangle.

**Solution:** Let  $A, B, C$  be the points  $u, v, \frac{u - iKv}{1 - iK}$  respectively on an Argand diagram.

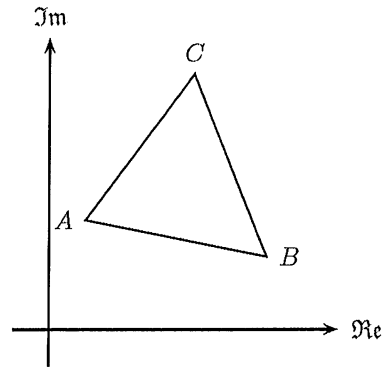
$$\overrightarrow{AB} = v - u.$$

$$\begin{aligned}\overrightarrow{BC} &= \frac{u - iKv}{1 - iK} - v, \\ &= \frac{u - iKv - v + iKv}{1 - iK}, \\ &= \frac{u - v}{1 - iK}.\end{aligned}$$

$$\begin{aligned}\overrightarrow{AC} &= \frac{u - iKv}{1 - iK} - u, \\ &= \frac{u - iKv - u + iKu}{1 - iK}, \\ &= \frac{iK(u - v)}{1 - iK}.\end{aligned}$$

$$\therefore \overrightarrow{AC} = iK \overrightarrow{BC}.$$

i.e.,  $\overrightarrow{AC}$  is rotated  $90^\circ$  from  $\overrightarrow{BC}$ .



- (c) The quadratic equation

$$x^2 - (2 \cos \theta)x + 1 = 0$$

has roots  $\alpha$  and  $\beta$ .

- (i) Find expressions for  $\alpha$  and  $\beta$  in terms of  $\theta$ .

**Solution:** Using the quadratic equation to find  $x$ ,

$$\begin{aligned}x &= \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2}, \\ &= \cos \theta \pm \sqrt{-(1 - \cos^2 \theta)}, \\ &= \cos \theta \pm i \sqrt{\sin^2 \theta}, \\ &= \cos \theta \pm i \sin \theta.\end{aligned}$$

$$\text{So } \alpha = \cos \theta + i \sin \theta,$$

$$\text{and } \beta = \cos \theta - i \sin \theta,$$

$$= \cos(-\theta) + i \sin(-\theta).$$

- (ii) Show that  $\alpha^{10} + \beta^{10} = 2 \cos(10\theta)$ .

$$\text{Solution: } \alpha^{10} = (\cos \theta + i \sin \theta)^{10},$$

$$= \cos 10\theta + i \sin 10\theta \text{ by deMoivre's theorem,}$$

$$\beta^{10} = (\cos(-\theta) + i \sin(-\theta))^{10},$$

$$= \cos(-10\theta) + i \sin(-10\theta) \text{ by deMoivre's theorem,}$$

$$\therefore \alpha^{10} + \beta^{10} = \cos 10\theta + i \sin 10\theta + \cos(-10\theta) + i \sin(-10\theta),$$

$$= \cos 10\theta + i \sin 10\theta + \cos 10\theta - i \sin 10\theta,$$

$$= 2 \cos(10\theta).$$

6. (a) When a cubic polynomial  $A(x)$  is divided by  $x^2 + x + 1$  the remainder is  $2x + 3$ . When  $A(x)$  is divided by  $x(x + 3)$  the remainder is  $5(x + 1)$ . Find the polynomial  $A(x)$ .

3

**Solution:**  $A(x) = (x^2 + x + 1)(ax + b) + 2x + 3,$   
 $= x(x + 3)(cx + d) + 5(x + 1),$

Put  $x = 0,$

$b + 3 = 5,$

$b = 2.$

Equating coefficients of  $x^3, a = c.$

Equating coefficients of  $x^2, a + 2 = 3c + d$  or  $2a = 2 - d.$

Equating coefficients of  $x, 2 + a + 2 = 3d + 5,$

$a = 3d + 1.$

Now  $2a = 2 - \frac{a - 1}{3},$

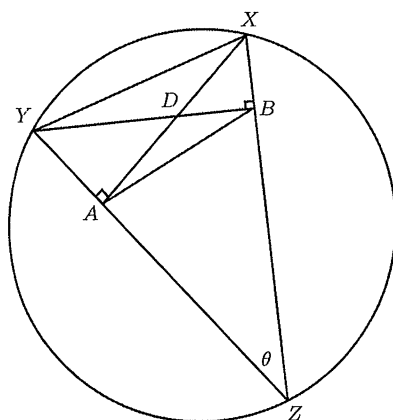
$6a = 6 - a + 1,$

$7a = 7,$

$a = c = 1.$

$\therefore A(x) = (x^2 + x + 1)(x + 2) + 2x + 3,$   
 $= x^3 + x^2 + x + 2x^2 + 2x + 2 + 2x + 3,$   
 $= x^3 + 3x^2 + 5x + 5.$

(b)



$XY$  is a fixed chord of a circle.  $Z$  is a point on the major arc  $XY$ .

The perpendicular from  $X$  to the chord  $YZ$  meets  $YZ$  at  $A$ .

The perpendicular from  $Y$  to the chord  $XZ$  meets  $XZ$  at  $B$ .

Let  $\angle XZY = \theta$ .

- (i) Copy the diagram into your booklet.

**Solution:** See above.

- (ii) Prove  $ABXY$  is a cyclic quadrilateral.

1

**Solution:**  $\angle XAY = 90^\circ$  ( $XA \perp YZ$ ),

$\angle YBX = 90^\circ$  ( $YB \perp XZ$ ),

$\therefore \angle XAY = \angle YBX.$

$\therefore ABXY$  is a cyclic quadrilateral (angles on arc  $XY$  are equal).

(iii) Show that  $AB = XY \cos \theta$ .

3

**Solution:**  $\angle ABZ = \angle XYZ$  (exterior  $\angle$  theorem for cyclic quad.)  
 $\angle BZA = \angle YZX$  (common),  
 $\therefore \triangle ABZ \sim \triangle XYZ$  (equiangular),  
 $\therefore \frac{AB}{XY} = \frac{ZA}{ZX}$  (corresp. sides of similar  $\triangle$ s),  
but  $\frac{ZA}{ZX} = \cos \theta$ ,  
 $\therefore \frac{AB}{XY} = \cos \theta$ ,  
so  $AB = XY \cos \theta$ .

(iv) Deduce that as  $Z$  moves on the major arc  $XY$  the length of  $AB$  is constant.

2

**Solution:**  $XY$  is constant (given),  
 $\theta$  is constant ( $\angle$ s at circumf. on same arc).  
 $\therefore \cos \theta$  is constant,  
 $\therefore AB = \text{constant} \times \text{constant}$ ,  
 $= \text{constant}$ .

(c) (i) Suppose that  $k$  is a double root of the polynomial equation  $P(x) = 0$ . Show that  $P'(k) = 0$ .

2

**Solution:** Put  $P(x) = (x - k)^2 Q(x) = 0$ , where  $Q(x)$  is a polynomial.  
 $P'(x) = 2(x - k)Q(x) + (x - k)^2 Q'(x)$ ,  
 $= (x - k)\{2Q(x) + (x - k)Q'(x)\}$ .  
 $\therefore P'(k) = (k - k)\{2Q(k) + (k - k)Q'(k)\}$ ,  
 $= 0$ .

(ii) If  $x^3 + 3px^2 + 3qx + r = 0$  has a double root, prove that  $(pq - r)^2 = 4(p^2 - q)(q^2 - pr)$ .

4

**Solution:** Let  $P(x) = x^3 + 3px^2 + 3qx + r = 0 \dots\dots\dots$  [1]  
 $P'(x) = 3x^2 + 6px + 3q = 0$ ,  
i.e.,  $x^2 + 2px + q = 0 \dots\dots\dots$  [2]  
 $x \times$  [2] :  $x^3 + 2px^2 + qx = 0 \dots\dots\dots$  [3]  
[1] - [3] :  $px^2 + 2qx + r = 0, \dots\dots\dots$  [4]  
 $p \times$  [2] :  $px^2 + 2p^2x + qp = 0, \dots\dots\dots$  [5]  
[5] - [4] :  $2(p^2 - q)x + qp - r = 0$ ,  
 $x = \frac{r - pq}{2(p^2 - q)}$ .  
Sub. in [2] :  $\frac{(r - pq)^2}{4(p^2 - q)^2} + \frac{2p(r - pq)}{2(p^2 - q)} + q = 0$ ,  
 $\frac{(pq - r)^2}{4(p^2 - q)^2} = \frac{p(pq - r)}{p^2 - q} - q$ ,  
 $(pq - r)^2 = 4(p^2 - q)(pq - r)p - 4q(p^2 - q)^2$ ,  
 $= 4(p^2 - q)(p^2q - pr - p^2q + q^2)$ ,  
 $= 4(p^2 - q)(q^2 - pr)$ .